

```

timescale 1ns / 1ps
`default_nettype none
`include "parameters.vh"

module top_module_tb;

    reg clk;
    reg rst_n;
    wire [31:0] ex_result;
    wire [31:0] if_instr;
    wire [31:0] if_Pc;
    wire [31:0] id_rs1_data;
    wire [31:0] id_rs2_data;
    wire [31:0] id_imm_data;
    wire [31:0] id_pc;
    wire [2:0] id_func3;
    wire id_regSrc, id_sel1, id_sel2, id_load, id_branch, id_jal, id_jalr, id_memSrc;
    wire [1:0] id_resultSrc;
    wire [3:0] id_alu_ctrl;
    wire [31:0] ex_pc, ex_imm_data, ex_data_store;
    wire [2:0] ex_func3;
    wire ex_load, ex_boj, ex_jalr, ex_regsrc, ex_memsrc;
    wire [1:0] ex_resultsrc;
    wire [31:0] result_to_if;
    wire [4:0] id_rd, ex_rd, m_rd, w_rd;
    wire m_regSrc, w_regsrc;
    wire [1:0] m_resultsrc;
    wire [31:0] m_result, m_data, m_pc4, w_data;
    wire [31:0] fwdwire1, fwdwire2;
    wire o_load_use_hazard;

    // Instantiate top_module
    top_module uut (
        .clk(clk),
        .rst_n(rst_n),
        .ex_result(ex_result),
        .if_instr(if_instr),
        .if_Pc(if_Pc),
        .id_rs1_data(id_rs1_data),
        .id_rs2_data(id_rs2_data),
        .id_imm_data(id_imm_data),
        .id_pc(id_pc),
        .id_func3(id_func3),
        .id_regSrc(id_regSrc),
        .id_sel1(id_sel1),
        .id_sel2(id_sel2),

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.id_load(id_load),
.id_branch(id_branch),
.id_jal(id_jal),
.id_jalr(id_jalr),
.id_memSrc(id_memSrc),
.id_resultSrc(id_resultSrc),
.id_alu_ctrl(id_alu_ctrl),
.ex_pc(ex_pc),
.ex_imm_data(ex_imm_data),
.ex_data_store(ex_data_store),
.ex_func3(ex_func3),
.ex_load(ex_load),
.result_to_if(result_to_if),
.ex_boj(ex_boj),
.ex_jalr(ex_jalr),
.ex_regsrc(ex_regsrc),
.ex_memsrc(ex_memsrc),
.ex_resultsrc(ex_resultsrc),
.m_rd(m_rd),
.m_regSrc(m_regSrc),
.m_resultsrc(m_resultsrc),
.m_result(m_result),
.m_data(m_data),
.m_pc4(m_pc4),
.w_regsrc(w_regsrc),
.w_data(w_data),
.id_rd(id_rd),
.ex_rd(ex_rd),
.w_rd(w_rd),
.fwdwire2(fwdwire2),
.fwdwire1(fwdwire1),
.o_load_use_hazard(o_load_use_hazard)
);

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// Clock generation
always #5 clk = ~clk;

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initial begin
    $dumpfile("top_module_tb.vcd");
    $dumpvars(0, top_module_tb);

    clk = 0;
    rst_n = 0;
    #20;
    rst_n = 1;

```

```
// Wait for a few hundred ns to let instructions execute
#1000;

$display("\nFinal ex_result: %h", ex_result);
$finish;
end

endmodule
```